Course Name: Blockchain and its Applications (NOC26_CS34)

Assignment 7 - Week 7 (Feb 2026)

TYPE OF QUESTION: MCQ/MSQ

Number of questions: **10** **Total mark: 10 X 1 = 10**

QUESTION 1

Which of the following properties make PBFT suitable for enterprise blockchain environments?

a. Deterministic transaction finality
b. Anonymous validator participation
c. Tolerance to Byzantine faults
d. Dependence on computational mining

Answer: (a) and (c)

Detailed solution:
PBFT provides deterministic finality - once a block is committed, it cannot be reverted. It tolerates Byzantine (arbitrary or malicious) faults. It does not rely on mining and assumes known validator identities, making it suitable for permissioned enterprise systems.

QUESTION 2

In Hyperledger Fabric, what is the primary role of the Membership Service Provider (MSP)?

a. Mining new blocks
b. Defining the rules for identity validation and managing the roles of participants.
c. Executing smart contracts
d. Ordering transactions

Answer: (b)

Detailed solution:
The Membership Service Provider (MSP) is basically the piece that handles identity management in the network. The Certificate Authority (CA) is the one that actually issues certificates, but the MSP decides which CAs are trusted and how identities are interpreted. It connects a given certificate to a specific role in an organization or channel, like identifying someone as an “admin” or a “peer.” In short, it’s what turns a regular cryptographic certificate into a recognized member of the network.

QUESTION 3

Which of the following conditions must hold in a Byzantine quorum system to ensure safety?

a. Any two quorums must intersect in at least one correct (honest) replica
b. Quorums may be completely disjoint to increase throughput
c. Quorum size must equal exactly half of the network
d. Faulty replicas must be able to form a standalone quorum.

Answer: (a)

Detailed solution:
Safety in a Byzantine environment comes from the Quorum Intersection Property. To avoid "split-brain" situations where two different values get committed, any two quorums have to overlap. That overlap needs to include at least $f + 1$ replicas. Since the system can have at most f faulty replicas, having $f + 1$ in common guarantees that at least one honest replica is part of both quorums, which preserves consistency.

QUESTION 4

Which of the following are advantages of blockchain in supply-chain systems?

a. Tamper-evident audit trail
b. Automatic deletion of transaction history
c. Improved transparency among participants
d. Removal of all governance requirements

Answer: (a) and (c)

Detailed solution:
Blockchain provides immutability and traceability, creating tamper-evident audit trails. It enhances transparency among stakeholders. However, governance is still required, and historical records are not automatically deleted.

QUESTION 5

What mechanism ensures progress in PBFT when the primary replica becomes faulty?

a. Increasing block size
b. Timeout-based view change protocol
c. Switching to PoW
d. Re-broadcasting old transactions

Answer: (b)

Detailed solution:
PBFT includes a view change protocol. When replicas detect that the primary is faulty or unresponsive (via timeout), they initiate a view change to elect a new primary, ensuring liveness.

QUESTION 6

Which of the following statements correctly describe permissioned blockchain systems?

a. Participants are authenticated before joining
b. All nodes remain anonymous
c. Access to transaction data may be restricted
d. Consensus mechanisms are unnecessary

Answer: (a) and (c)

Detailed solution:
Permissioned blockchains require identity verification before participation. They can restrict transaction visibility to certain participants. However, consensus mechanisms are still required to maintain agreement.

QUESTION 7

In Hyperledger Fabric, what is the purpose of channels?

a. To increase mining difficulty
b. To create logically separate ledgers among subsets of participants
c. To replace consensus mechanisms
d. To store private cryptographic keys

Answer: (b)

Detailed solution:
Channels allow a subset of organizations to maintain a private ledger visible only to channel members. This ensures confidentiality within a broader network.

QUESTION 8

In a Byzantine Fault Tolerant (BFT) system with f malicious nodes, why is the total number of replicas $n \geq 3f + 1$ required?

a. To ensure that any two quorums of size $2f + 1$ share at least one honest replica.
b. To eliminate digital signatures and hashing.
c. To ensure the system can function with $2f + 1$ faulty nodes.
d. To minimize the number of communication rounds to *O(1)*

Answer: (a)

Detailed solution:
If there are f Byzantine faulty nodes, at least $3f + 1$ total replicas are required so that decision quorums contain more honest nodes than faulty ones, preserving safety and agreement.

QUESTION 9

Which of the following properties characterize committee-based consensus protocols such as Algorand?

a. Committee membership is rotated and unique for every round
b. Selection is based on cryptographic randomness (VRF)
c. Membership is publicly predictable long in advance
d. The entire network validates every block to ensure maximum security.

Answer: (a) and (b)

Detailed solution:
Algorand uses cryptographic sortition (via VRFs) to randomly select committees. Membership changes each round and is not predictable in advance, enhancing security.

QUESTION 10

Which of the following are key architectural features of Hyperledger Fabric?

a. Modular design with pluggable components
b. Mandatory native cryptocurrency
c. Separation of transaction endorsement and ordering
d. Public anonymous participation

Answer: (a) and (c)

Detailed solution:
Fabric has a modular architecture allowing pluggable components such as consensus mechanisms. It separates transaction endorsement from ordering, improving flexibility. It does not require a native cryptocurrency and is permissioned rather than anonymous.